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JEE ADVANCED



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A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

SEC: Jr.Super60_NUCLEUS & STERLING BT CTA-13/CTA-02_(QAT-02)
Time: 02:00PM to 05:00PM JEE-ADV_(2020-P1)

Date: 29-10-2023
Max. Marks: 198

2020_PAPER-I

**29-10-2023_Jr.Super60_NUCLEUS & STERLING BT_Jee-Adv
(2020-P1)_CTA-13&CTA-02_(QAT-02)_Syllabus**

PHYSICS : TOTAL SYLLABUS COVERED FROM 23-APR-2023 TO 27-OCT-2023
CHEMISTRY : TOTAL SYLLABUS COVERED FROM 23-APR-2023 TO 29-OCT-2023
MATHEMATICS : SYLLABUS COVERED FROM 01-MAY-2023 TO 18-OCT-2023

Name of the Student: _____

H.T. NO:

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**JEE-ADVANCE-2020-P1-Model**

Time: 3:00Hour's

IMPORTANT INSTRUCTIONS**Max Marks: 198****PHYSICS:**

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total mark
Sec – I(Q.N : 1 – 6)	Questions with Single Correct Choice	3	-1	6	18
Sec – II(Q.N : 7 – 12)	Questions with Multiple Correct Choice +1 partial marks	4	-2	6	24
Sec – III(Q.N : 13 – 18)	Questions with Numerical Value Answer Type	4	0	6	24
Total				18	66

CHEMISTRY:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 19 – 24)	Questions with Single Correct Choice	3	-1	6	18
Sec – II(Q.N : 25 – 30)	Questions with Multiple Correct Choice +1 partial marks	4	-2	6	24
Sec – III(Q.N : 31 – 36)	Questions with Numerical Value Answer Type	4	0	6	24
Total				18	66

MATHEMATICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 37 – 42)	Questions with Single Correct Choice	3	-1	6	18
Sec – II(Q.N : 43 – 48)	Questions with Multiple Correct Choice +1 partial marks	4	-2	6	24
Sec – III(Q.N : 49 – 54)	Questions with Numerical Value Answer Type	4	0	6	24
Total				18	66

Sec: Jr.Super60_NUCLEUS & STERLING BT

Space for rough work

Page 2

**Sri Chaitanya**
Educational Institutions**THE PERFECT HAT-TRICK WITH ALL- INDIA RANK 1
IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023****JEE MAIN
2023**SINGARAJU
VENKAT KOUNDINYA
AIR-1000000
Sri Chaitanya
B-12th Class**300
300****RANK
1****JEE Advanced
2023**VARILALA
CHIDVILAS REDDY
AIR-1000000
Sri Chaitanya
B-12th Class**341
360****RANK
1****NEET
2023**BORA VARUN
CHAKRAVARTHI
AIR-1000000
Sri Chaitanya
B-12th Class**720
720****RANK
1**

PHYSICS

Max Marks: 66

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains **SIX** (06) questions.

• Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.

• For each question, choose the option corresponding to the correct answer.

• Answer to each question will be evaluated **according to the following marking scheme:**

Full Marks : +3 If **ONLY** the correct option is chosen;

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);

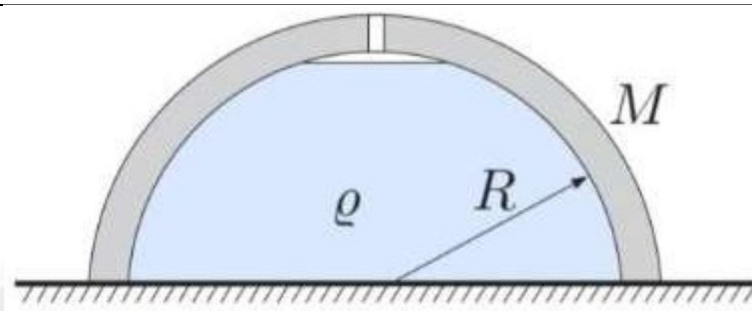
Negative Marks : -1 In all other cases

- One mole of an ideal gas is contained under a weightless conducting piston in a vertical cylinder at a temperature T . The space over the piston is open to the atmosphere. The amount of work that has to be done to increase the gas volume in the cylinder to n times by slowly raising the piston is (Neglect friction)
A) $RT[(n-1)-\ln(n)]$ B) $\frac{RT}{2}[(n-1)-\ln(n)]$ C) $\frac{RT}{2}\left[(n-1)-\frac{\ln(n)}{2}\right]$ D) $RT\left[(n-1)-\frac{\ln(n)}{2}\right]$
- The pressure of one mole of an ideal gas varies according to the equation $p=p_0-av^2$, where p_0 and a are positive constants. The maximum temperature attained by the gas during the process is
A) $\frac{3p_0}{2R}\left(\frac{p_0}{3a}\right)^{\frac{1}{2}}$ B) $\frac{p_0}{R}\left(\frac{p_0}{3a}\right)^{\frac{1}{2}}$ C) $\frac{2p_0}{R}\left(\frac{p_0}{3a}\right)^{\frac{1}{2}}$ D) $\frac{2p_0}{3R}\left(\frac{p_0}{3a}\right)^{\frac{1}{2}}$
- Power radiated by a perfectly black body is p_0 and wavelength corresponding to maximum energy is λ_0 . On changing temperature the wavelength corresponding to maximum energy is increased by $\frac{\lambda_0}{2}$. Now the power radiated by the body will be
A) $\frac{2p_0}{3}$ B) $\frac{4p_0}{9}$ C) $\frac{16p_0}{81}$ D) $\frac{256p_0}{729}$
- A hemispherical container is placed upside down on a smooth horizontal surface. Through a small hole at the bottom of the container, water is then poured in. Exactly when the container gets full, water starts leaking from between the table and the edge of the container. Find the mass of the container if water has density ρ and radius of the hemisphere is R .

Sec: Jr.Super60_NUCLEUS & STERLING BT

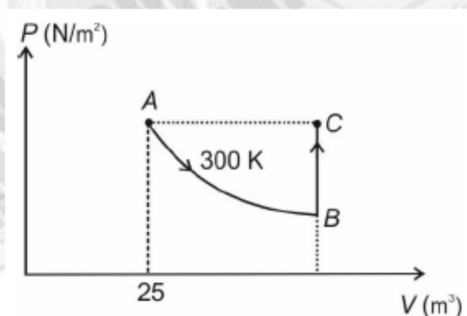
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Page 3



- A) $\frac{2}{3}\pi R^3 \rho$ B) $\pi R^3 \rho$ C) $\frac{1}{3}\pi R^3 \rho$ D) $\frac{4}{3}\pi R^3 \rho$

5. One mole of an ideal monatomic gas initially at temperature 300 K is isothermally expanded to four times its initial volume and then isochorically heated so that the pressure in the final state becomes equal to its initial value. The process is shown in the P-V graph (Take, $R = \frac{25}{3} \text{ J mol}^{-1} \text{ K}^{-1}$)



- A) Slope of the curve $A \rightarrow B$ when volume is 50 m^3 is -5 in SI units
 B) Slope of the curve $A \rightarrow B$ when volume is 50 m^3 is -1 in SI units
 C) Heat absorbed in the process $A \rightarrow B \rightarrow C$ is nearly 11.25 KJ
 D) Heat absorbed in the process $A \rightarrow B \rightarrow C$ is nearly 10.7 KJ
6. Consider 2 moles of O_2 at 27°C and 4 moles of He at 127°C which are mixed with each other inside an insulated rigid container (assume no heat absorbed by the container). If T_{mix} and γ_{mix} , are the final temperature and adiabatic exponent of this mixture then we have (assume ideal gases)

- A) $T_{\text{mix}} = \frac{3900}{11} \text{ K}$ B) $\gamma_{\text{mix}} = \frac{11}{6}$ C) $\gamma_{\text{mix}} = \frac{20}{11}$ D) $\gamma_{\text{mix}} = \frac{800}{11} \text{ K}$





SECTION – II

(ONE OR MORE CORRECT ANSWER TYPE)

- This section contains **SIX** (06) questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated **according to the following marking scheme**:

Full Marks : +4 If only (all) the correct option(s) is(are) chosen;

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen;

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct;

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option;

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);

Negative Marks : -2 In all other cases

7. Two friends A and B are standing at a distance x apart in an open field and wind is blowing at a constant velocity u from A to B. A beats a drum and B hears its sound time t_1 after he sees the event. A and B interchange their position and A beats the drum again. This time B hears the drum time t_2 after he sees the event. Neglect the time light takes in travelling between the friends. Wind is continuously blowing in the same direction. Choose the correct options.

A) Velocity of sound in still air, $v = x \left(\frac{1}{t_1} + \frac{1}{t_2} \right)$

B) Velocity of wind, $u = \frac{x}{2} \left(\frac{1}{t_1} - \frac{1}{t_2} \right)$

C) Velocity of sound in still air, $v = \frac{x}{2} \left(\frac{1}{t_1} + \frac{1}{t_2} \right)$

D) Velocity of wind, $u = x \left(\frac{1}{t_1} - \frac{1}{t_2} \right)$

8. A syringe is filled with water. Its volume is $v_0 = 40 \text{ cm}^3$ and cross section of its interior is $A = 8 \text{ cm}^2$. The syringe is held vertically such that its nozzle is at the top, and its piston is pushed up with constant speed. Mass of the piston is $M = 50 \text{ g}$ and the water is ejected at a speed of $u_0 = 2 \text{ m/s}$. Cross section of the stream of water at the nozzle is $a = 2 \text{ mm}^2$. Neglect friction, take the density of water to be $\rho = 10^3 \text{ kg/m}^3$, $g = 10 \text{ m/s}^2$. Choose the correct options.



**JEE MAIN
2023**

SINGARAJU
VENKAT KOUNDINYA
AIR-1000000
Sri Chaitanya
6th-12th Class
**300
300**
RANK



**RANK
1**

**JEE Advanced
2023**

VARILALA
CHIVILAS REDDY
AIR-1000000
Sri Chaitanya
6th-12th Class
**341
360**
RANK



**RANK
1**

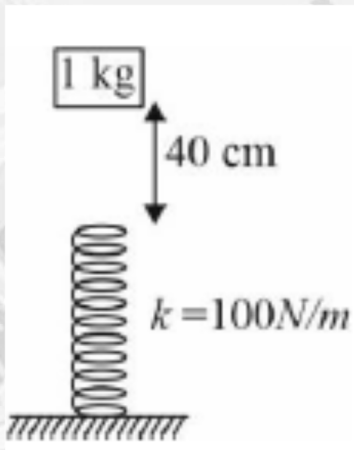
**NEET
2023**

BORA VARUN
CHAKRAVARTHI
AIR-1000000
Sri Chaitanya
6th-12th Class
**720
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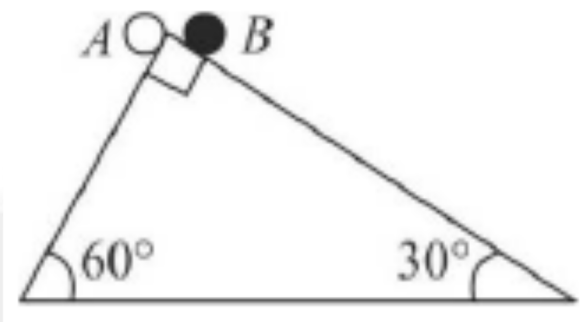


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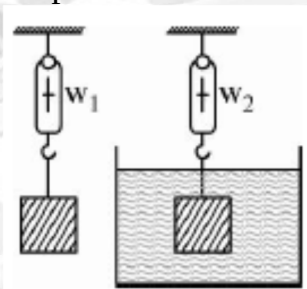
- A) The speed of the piston is 0.5 cm/s
B) The force required to push the piston decreases as the piston moves up
C) Total work done in emptying the syringe is nearly 0.115 J
D) Total work done in emptying the syringe is nearly 0.125 J
9. A long spring is kept vertical with its lower end attached to the ground and the upper end free. A block is released above the spring so that it strikes the top of spring after falling through a distance of 40 cm as shown. As the block compresses the spring, assume that the spring remains always vertical. Mass of the block is 1 kg and spring constant is 100 N/m. Choose the correct options. (Take $g=10\text{m/s}^2$).
- 
- A) Velocity of the block is maximum just before striking the spring
B) Velocity of the block is maximum when the compression in spring is 10 cm
C) Maximum compression of the spring is 10 cm
D) Maximum compression of the spring is 40 cm
10. A hollow sphere A and a solid sphere B, both having same mass and radius are released simultaneously from the top of a right triangular fixed wedge as shown. They roll down the wedge without slipping. Choose the correct statements.





- A) They reach the bottom of the wedge simultaneously
- B) A reaches the bottom of wedge before B
- C) They reach the bottom of wedge with the same kinetic energy
- D) A reaches the bottom of wedge with speed more than that of B

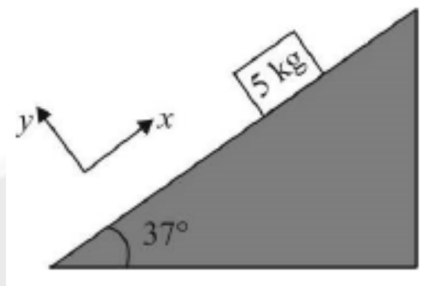
11. A spring balance reads w_1 with a block suspended from it. If the block is submerged in a liquid whose density is less than density of block, the spring balance reads w_2 . The experiment is repeated twice, once on surface of earth and another time on surface of moon. If acceleration due to gravity on surface of moon is one-sixth of that on surface of earth, then choose the correct options.



- A) w_1 is same on earth and moon
- B) w_2 is same on earth and moon
- C) $(w_1 - w_2)$ is same on earth and moon
- D) $\frac{w_1}{w_2}$ is same on earth and moon

12. A block of mass 5 kg is kept on a fixed incline. The coefficient of friction between the block and incline is 0.5. x and y axis as shown are parallel to the incline and perpendicular to the incline respectively. Choose the correct options.

(Take $g = 10\text{m/s}^2$, $\tan 37^\circ = 0.75$)



- A) If the block is released, it slides down the incline with acceleration 6m/s^2
 B) If the block is released, it slides down the incline with acceleration 2m/s^2
 C) The minimum magnitude of force that needs to be applied on the block along $+x$ direction to keep the block at rest is 10N
 D) The minimum magnitude of force that needs to be applied on the block along $-y$ direction to keep the block at rest is 20N

SECTION – III (NUMERICAL VALUE TYPE)

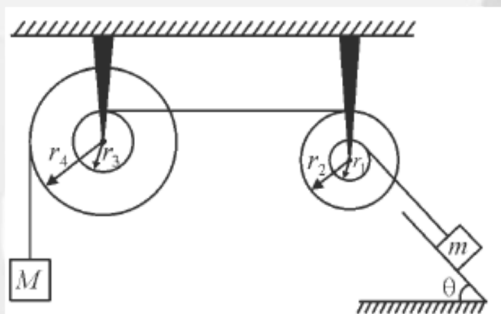
- This section contains **SIX** (06) questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer. **If** the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.
- Answer to each question will be evaluated **according to the following marking scheme**:
Full Marks : +4 **If ONLY** the correct numerical value is entered;
Zero Marks : 0 In all other cases

13. Starting from rest, a particle moves a distance x in a straight line with constant acceleration a . Then, the particle moves a further distance x along the same line at constant velocity. Finally, the particle, moving along the same line, decelerates uniformly at rate $\frac{a}{2}$ until it comes to rest. Let the maximum velocity of the particle during its motion be v_{max} and let its average velocity over the entire journey be v_{AVG} . The ratio $\frac{v_{\text{MAX}}}{v_{\text{AVG}}}$ is equal to _____.
14. A ball initially at rest is given a sharp horizontal impulse at a height h above the centre of the ball. As a result, the ball immediately begins to roll without slipping. If the ball is a solid sphere, the height $h=h_1$ while if the ball is a hollow sphere, the height $h=h_2$. Find $\frac{h_1}{h_2}$





15. A thin hollow spherical shell of radius R is cut symmetrically and its lower half is removed to get a hemispherical shell. Now the open base is covered with a circular plate of radius R made of same material and having the same thickness as the shell. The centre of mass of this covered hemisphere lies at a distance of nR from the centre of base, where n is _____.
16. In the arrangement shown, the strings are mass less, and the radii of the pulleys are: $r_1=3\text{cm}, r_2=6\text{cm}, r_3=4\text{cm}$ and $r_4=10\text{cm}$. Friction is absent everywhere and the angle $\theta=\sin^{-1}\left(\frac{4}{5}\right)$. The value of $\frac{M}{m}$ such that the system remains in equilibrium is _____.



17. From a point on the ground at distance of 3m from the foot of a vertical wall, a ball is thrown at an angle 60° . The ball just clears the top of the wall and afterwards strikes the ground at a distance of 6m from the foot of the wall on the other side. The height of the wall is _____ (in m). (Take $\sqrt{3}=1.73$)
18. A block of mass 1 kg is kept at rest on a rough horizontal surface. The coefficient of friction between the block and surface is 0.5 . A horizontal force of 10 N is applied on the block for 6 seconds after which the direction of force is reversed keeping the magnitude same. Find the speed (in m/s) with which the block returns to its starting point. (Take $g = 10\text{m/s}^2, \sqrt{3} = 1.732$)

Sri Chaitanya Educational Institutions
INDIA



CHEMISTRY

Max Marks: 66

SECTION – I
(SINGLE CORRECT ANSWER TYPE)This section contains **SIX** (06) questions.• Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.

• For each question, choose the option corresponding to the correct answer.

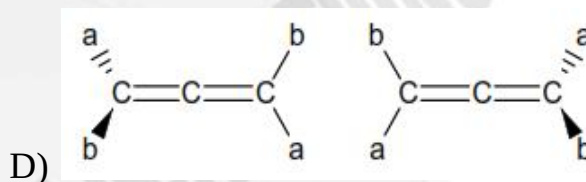
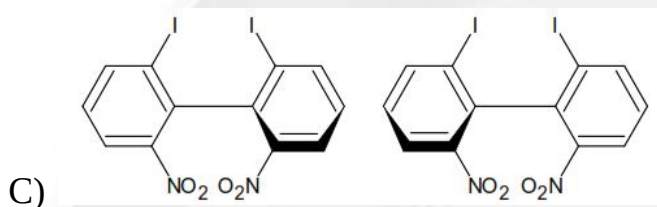
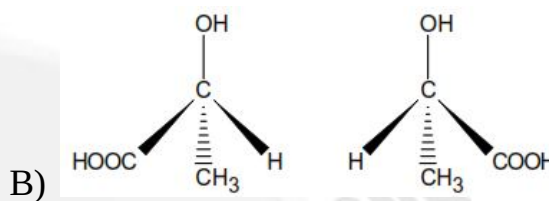
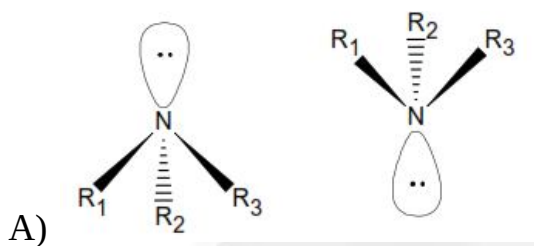
• Answer to each question will be evaluated **according to the following marking scheme:****Full Marks** : +3 If **ONLY** the correct option is chosen;**Zero Marks** : 0 If none of the options is chosen (i.e. the question is unanswered);**Negative Marks** : -1 In all other cases

19. An electron, practically at rest is initially accelerated through a potential of 100volts. It then has a de Broglie wavelength = $\lambda_1 \text{ \AA}$. It then gets retarded through 19 volts and then has a wavelength = $\lambda_2 \text{ \AA}$, further retardation through 32 volts changes the wavelength, now the wavelength = $\lambda_3 \text{ \AA}$. What is the value of $\frac{\lambda_3 - \lambda_2}{\lambda_1}$?
- A) $\frac{20}{41}$ B) $\frac{10}{63}$ C) $\frac{20}{63}$ D) $\frac{10}{41}$
20. The normal boiling point of a liquid X is 400K. ΔH_{vap} at normal boiling point is 40kJ/mol select correct statement:
- A) $\Delta S_{\text{vaporisation}} < 100 \text{ J / mol.K}$ at 400K and 2 atm
- B) $\Delta S_{\text{vaporisation}} < 100 \text{ J / mol.K}$ at 400K and 1 atm
- C) $\Delta G_{\text{vaporisation}} > 0$ at 400K and 1 atm
- D) $\Delta U = 93.32 \text{ kJ / mol.K}$ at 400K and 1 atm
21. Which of the following are polar molecules?
- 1) NH_3 2) AsF_3 3) CHCl_3 4) BeF_2 5) SnCl_4
- A) 1,2 and 3 only B) 1 and 3 only C) 2,3 and 4 only D) 2,4 and 5 only





22. Which of the following pairs cannot be resolved



23. The 100ml of 0.1M CH_3COOH ($K_a = 10^{-5}$) is titrated with 0.1 M NaOH and pH is observed at three states at 298K.

1st step $\Rightarrow$ 50ml of 0.1M NaOH added.

2nd step $\Rightarrow$ 100ml of 0.1 M NaOH added.

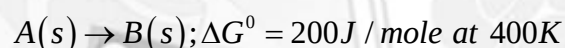
3rd step $\Rightarrow$ 120ml of 0.1 M NaOH added.

Calculate the difference in pH between stages (I & II) and (II & III) stages.

A) $(\text{pH}_{\text{II}} - \text{pH}_{\text{I}}) = 3.84$ & $(\text{pH}_{\text{III}} - \text{pH}_{\text{II}}) = 4.11$ B) $(\text{pH}_{\text{I}} - \text{pH}_{\text{II}}) = 2$ & $(\text{pH}_{\text{III}} - \text{pH}_{\text{II}}) = 5.2$

C) $(\text{pH}_{\text{II}} - \text{pH}_{\text{I}}) = 3.84$ & $(\text{pH}_{\text{III}} - \text{pH}_{\text{II}}) = 3.11$ D) $(\text{pH}_{\text{I}} - \text{pH}_{\text{II}}) = 4.2$ & $(\text{pH}_{\text{III}} - \text{pH}_{\text{II}}) = 3.11$

24. For an allotropic change:



If molar volumes (independent of P) of A and B are 12 L /mole and 10 L/mole respectively then calculate approximate pressure at which the two allotropes will be at equilibrium at 400K:

A) 10^5 Pa B) $2 \times 10^5 \text{ Pa}$ C) 100 Pa D) 10^{10} Pa

SECTION – II (ONE OR MORE CORRECT ANSWER TYPE)

- This section contains **SIX** (06) questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).





• Answer to each question will be evaluated according to the following marking scheme:

Full Marks	: +4 If only (all) the correct option(s) is(are) chosen;
Partial Marks	: +3 If all the four options are correct but ONLY three options are chosen;
Partial Marks	: +2 If three or more options are correct but ONLY two options are chosen, both of which are correct;
Partial Marks	: +1 If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks	: 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	: -2 In all other cases

25. Which of the following statement(s) is/are correct?
- A) BeF_2 from complex ion with NaF in which Be goes with cation
- B) LiCl is soluble in pyridine
- C) $Be(OH)_2$ dissolves in alkali forming $[Be(OH)_4]^{2-}$
- D) One of the ingredient present in Portland cement is $Ca_3Al_2O_6$
26. When a equimolar mixture of Cu_2S and CuS is titrated with $Ba(MnO_4)_2$ in acidic medium, the final products contain Cu^{2+} , SO_2 and Mn^{2+} . If the mol. mass of Cu_2S , CuS and $Ba(MnO_4)_2$ are M_1 , M_2 and M_3 respectively then:
- A) eq. mass of Cu_2S is $\frac{M_1}{8}$
- B) eq mass of CuS is $\frac{M_2}{6}$
- C) eq. mass of $Ba(MnO_4)_2$ is $\frac{M_3}{5}$
- D) Cu_2S and CuS both have same equivalents in mixture.
27. Choose the correct options:
- A) Favourable conditions for formation of graphite are high pressure and low temperature from equilibrium diamond ($d=3.5$ g/ml) $\rightleftharpoons$ graphite ($d=2.3$ g/ml); $\Delta H = -1.9$ kJ / mole
- B) For reaction $N_2O_4(g) \rightleftharpoons 2NO_2(g)$; degree of dissociation (α) is $\sqrt{\frac{K_p}{4P + K_p}}$ where P is equilibrium pressure
- C) For reaction $Cl_2(g) + 3F_2(g) \rightleftharpoons 2ClF_3(g)$; $\Delta H = -329$ kJ, dissociation of $ClF_3(g)$ will be favoured by addition of inert gas at constant pressure
- D) Solubility of gas in liquid increases with increasing of temperature.





28. Which of the following names are correct for the compound containing molecular formula is C_7H_8O ?
 A) Phenylmethanol B) 2-methylphenol C) 3-methylphenol D) methoxybenzene
29. The standard electrode potentials of some of the alkali metals are
 $Li^+/Li = -3.05V$; $Na^+/Na = -2.71V$; $K^+/K = -2.93V$
 Alkali metals react with water according to the following reaction. (M = alkali metal)
 $2M + H_2O \rightarrow 2MOH + H_2$
 It is found that Li reacts gently with water whereas potassium reacts violently with water. Which of the following statement(s) is/are correct regarding the above two experimental fact?
 A) ΔG values have nothing to do with the kinetics of a reaction
 B) The metal having large standard oxidation potential will have lesser reactivity
 C) K has a low melting point, and the heat of reaction is sufficient to make it melt or even vapourize. This leads to exponential increase in its surface area
 D) The ionization potential of K is greater than that of Li.
30. Which of the following statement is/are correct?
 A) O_2 is paramagnetic, N_2 is also paramagnetic
 B) O_2 is paramagnetic, N_2 is diamagnetic
 C) O_2 is paramagnetic, O_2^{2-} is also paramagnetic
 D) In $O_2 \rightarrow O_2^+$ and $NO \rightarrow NO^+$, bond length decreases.

SECTION – III

(NUMERICAL VALUE TYPE)

- This section contains **SIX** (06) questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer. **If** the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.
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Zero Marks : 0 In all other cases

31. The ratio of rate of diffusion of gases 'A' and 'B' is 1:4. If the ratio of their masses presenting the initial mixture is 2:3. The ratio of their mole fraction is 'p'. The value of '3p' is [use $(0.375)^{3/2} = 0.52$]
32. 0.9 gm of a volatile solid organic compound (molecular weight = 90) containing carbon, hydrogen and oxygen was heated with 224 ml of oxygen at 1 atm & $0^\circ C$. After combustion, the total volume of gases was 560 ml at same T and P. On treatment with aq.KOH, the volume decreased to 112 ml. Determine the value of $x + y + z$ if molecular formula of organic compound is $C_xH_yO_z$.

JEE MAIN
2023SINGARAJU
VENKAT KOUNDINYA
AIR-100
Sri Chaitanya
Jr-12th Class300
300
RANK

RANK

1

JEE Advanced
2023VARILALA
CHIVILAS REDDY
AIR-100
Sri Chaitanya
Jr-12th Class341
360
RANK

RANK

1

NEET
2023BORA VARUN
CHAKRAVARTHI
AIR-100
Sri Chaitanya
Jr-12th Class720
720
RANK

RANK

1



33. Bond distance C-F in (CF_4) & Si-F in (SiF_4) are respectively $1.33 \text{ \AA}$ & $1.54 \text{ \AA}$. C-Si bond is $1.87 \text{ \AA}$. Calculate the covalent radius of F atom ignoring the electronegativity differences.
34. The solubility of solid AgCN is affected by pH of solution due to hydrolysis of CN^{-1} ion. $AgCN(s) \rightleftharpoons Ag^+(aq) + CN^-(aq)$
 $CN^-(s) + H_2O \rightleftharpoons HCN + OH^-$
 Suppose the solubility of AgCN is $(x) \times 10^{-7}$ at $[H^+] = 3 \times 10^{-10} M$. Find the value of $(y-x)$. Given: $K_{sp}(AgCN) = 10^{-16}$
 $K_a(HCN) = 10^{-10}$
35. How many of the following orders/statements are correct.
- IP_1 of nitrogen is not less than that of oxygen
 - IP_1 of Pb is less than that of Sn
 - Electronegativity order $B > Tl > In > Ga > Al$
 - IP_2 order $B > C > Be$
 - Electron gain enthalpy order $Ne > Ar > Kr > Xe > He > Rn$
 - Electron affinity of Be^{+4} is 217.6 eV/ion nearly.
 - Electronegativities of K & Rb are same
 - Order of sizes $Tl > In > Al > Ga > B$
 - Official IUPAC name of element with atomic number 104 is Unnilquadium.
36. A 2:3 racemic mixture (by weight) has an observed specific rotation of 16° . The specific rotation of the optically pure component is $x \times 10$ degree where x is



MATHEMATICS

Max Marks: 66

**SECTION – I
(SINGLE CORRECT ANSWER TYPE)**

This section contains **SIX** (06) questions.

• Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.

• For each question, choose the option corresponding to the correct answer.

• Answer to each question will be evaluated **according to the following marking scheme:**

Full Marks : +3 If **ONLY** the correct option is chosen;

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);

Negative Marks : -1 In all other cases

37. Let $x_1 = \sqrt{2}, x_2 = \sqrt{2 + \sqrt{2}}, x_3 = \sqrt{2 + \sqrt{2 + \sqrt{2}}}, \dots, x_n = \sqrt{2 + \sqrt{2 + \sqrt{2 + \dots n \text{ times}}}}$. Then

$2 \sin \left(\left(1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots + \frac{1}{2^{2018}} \right) \frac{\pi}{4} \right)$ must be

A) x_{2020}

B) x_{2019}

C) x_{2018}

D) x_{2021}

38. In a ΔABC , $A = (1, \sqrt{3})$, $B = (0, 0)$, Incentre $I = \left(1, \frac{1}{\sqrt{3}} \right)$, then $C =$

A) (5, 0)

B) (4, 0)

C) (0, 4)

D) (2, 0)

39. Let $a_1 = 1, a_2 = 1 + a_1, a_3 = 1 + a_1 a_2, \dots, a_{n+1} = 1 + a_1 a_2 \dots a_n$, then $\frac{1}{a_1} + \frac{1}{a_2} + \frac{1}{a_3} + \dots \text{upto } \infty$ is equal to

A) 1

B) 2

C) $\rightarrow \infty$

D) 3

40. Let $g : \mathbb{R} \rightarrow \mathbb{R}$ be defined as $g(x) = \frac{x^3 + e^{2x} - 1}{2}$. If $g(f(x)) = x$ where $f\left(\frac{e^4 + 7}{2}\right) = \alpha$, then the number of solution(s) of the equation $\|x - 2| - 3| - \alpha = 0$ is (are)

A) 1

B) 2

C) 3

D) 4

41. Let $f(x, y) = \sin(x - 2y) \cos 2y + \cos(x - 2y) \sin 2y \forall x, y \in \mathbb{R}$, then the value of

$\lim_{x \rightarrow 0} \left[\frac{f(x, y)}{\sin x} + \left(1 - x + \frac{f(x, y)}{\cos x} \right)^{\frac{1}{x^2 f(x, y)}} \right]$ is equal to (where $[\cdot]$ denotes greatest integer function)

A) $e^{1/3}$

B) $e^{1/3} + 1$

C) $\frac{4}{3}$

D) 0

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Space for rough work

Page 15



42. Let $x, y \in \mathbb{R}$, $x > 0$, $y > 0$ satisfying the system of equations $x^{\log_y x} \cdot y = x^{\frac{3}{2}}$ and $\log_4 y \cdot \log_y (y - 3x) = 1$. Then which of the following is/are true
- A) Value of x satisfying system of equations is rational
 B) Value of x satisfying system of equations is irrational
 C) No values of x and y satisfy the system of equations
 D) Exactly one pair of (x, y) satisfy the system of equations

SECTION - II (ONE OR MORE CORRECT ANSWER TYPE)

- This section contains **SIX** (06) questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated **according to the following marking scheme**:

Full Marks	: +4 If only (all) the correct option(s) is(are) chosen;
Partial Marks	: +3 If all the four options are correct but ONLY three options are chosen;
Partial Marks	: +2 If three or more options are correct but ONLY two options are chosen, both of which are correct;
Partial Marks	: +1 If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks	: 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	: -2 In all other cases

43. Let $f(x) = \begin{cases} \frac{\ln(1+2x)}{x} & , -\frac{1}{2} < x < 0 \\ 2\cos x & , x = 0 \\ \frac{e^{2x}-1}{x} & , 0 < x < 1 \\ e^2-1 & , x \geq 1 \end{cases}$ then which of the following option (s) is/are correct?

- (A) $f(x)$ is continuous at $x = 0$ (B) $f(x)$ is not differentiable at $x = 0$
 (C) $f(x)$ is continuous at $x = 1$ (D) $\lim_{x \rightarrow 0^+} [f(x)] = 1$ (Where $[.]$ denotes GIF)

44. If $f(x) = \tan^{-1}\left(\frac{\sqrt{3}x-3x}{3\sqrt{3}+x^2}\right) + \tan^{-1}\left(\frac{x}{\sqrt{3}}\right)$, $0 \leq x \leq 3$, then which of the following option (s) is/are correct?

- (A) the least value of $f(x)$ is $-\frac{\pi}{3}$ (B) the greatest value of $f(x)$ is $\frac{\pi}{4}$
 (C) the least value of $f(x)$ is 0 (D) the greatest value of $f(x)$ is $\frac{\pi}{3}$





45.

If α, β^2 are roots of $5x^2 - 20x + 3k = 0$ and β, α^2 are roots of $kx^2 - 50x + 45 = 0$

(a, b, $\alpha, \beta \in \mathbb{R}$ and $k \neq 0, \alpha > \beta$), then which of the following option (s) is/are correct?

(A) $\alpha\beta = 3$ (B) $k = 5$ (C) $\alpha^3 - 10\alpha + 3\beta = 0$ (D) $\beta^3 - 4\beta + \alpha = 0$

46.

Let $9 + 99 + 999 + \dots$ Upto n terms $= a$, then the correct option(s) is/are

(A) The value of $7 + 77 + 777 + \dots$ Upto n terms $= \frac{7a}{9}$

(B) The value of $7 + 77 + 777 + \dots$ Upto n terms $= \frac{9a}{7}$

(C) The value of $2^2 + 22^2 + 222^2 + \dots +$ upto n terms $= \frac{4}{81} \left[\frac{100(10^{2n} - 1)}{99} - (n + 2a) \right]$

(D) The value of $2^2 + 22^2 + 222^2 + \dots$ up to n terms $= \frac{4}{9} \left[\frac{100(10^{2n} - 1)}{99} - (n + 2a) \right]$

47.

let $P = \begin{bmatrix} 3 & -1 & -2 \\ 2 & 0 & \alpha \\ 3 & -5 & 0 \end{bmatrix}$, where $\alpha \in \mathbb{R}$.

Suppose $Q = [q_{ij}]$ is a matrix such that $PQ = KI$ where $K \in \mathbb{R}$, $K \neq 0$ and I is the identity matrix of order 3. If $q_{23} = -\frac{K}{8}$ and $\det(Q) = \frac{K^2}{2}$ then which of the following option (s) is/are correct?

(A) $\alpha = 0, K = 8$ (B) $4\alpha - K + 8 = 0$

(C) $\det(P \operatorname{adj}(Q)) = 2^9$ (D) $\det(Q \operatorname{adj}(P)) = 2^{13}$

48

Consider the matrix function: $A(x) = \begin{bmatrix} \cos^{-1} x & \sin^{-1} x & \operatorname{cosec}^{-1} x \\ \sin^{-1} x & \sec^{-1} x & \tan^{-1} x \\ \operatorname{cosec}^{-1} x & \tan^{-1} x & \cot^{-1} x \end{bmatrix}$ and $B = A^{-1}$

Which of the following statement(s) is/are correct?

(A) $|A(x)|$ is a continuous function in its domain but not differentiable in its domain.

(B) If $b = |\operatorname{adj} B| + |\operatorname{adj} B^2| + |\operatorname{adj} B^3| + \dots \infty$, then maximum value of b is $\frac{256}{\pi^6 - 256}$

(C) $|A(x)|$ is a bounded function

(D) If $a = \det(B) + \det(B^2) + \det(B^3) + \dots \infty$ then minimum value of a equals $\frac{8}{\pi^3 - 8}$





SECTION – III

(NUMERICAL VALUE TYPE)

- This section contains **SIX** (06) questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer. **If** the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.
- Answer to each question will be evaluated **according to the following marking scheme**:
Full Marks : +4 **If ONLY** the correct numerical value is entered;
Zero Marks : 0 In all other cases

49. If $\log_4(x+2y) + \log_4(x-2y) = 1$, the minimum value of $|x| - |y|$ is $\sqrt{\lambda}$ then find λ .
50. Let $M_n = \{0.a_1a_2\ldots a_n \mid a_i = 0 \text{ or } 1, 1 \leq i \leq n-1, a_n = 1\}$ be a set of decimal fractions. T_n and S_n be the number and the sum of the elements in M_n respectively. If $\lim_{n \rightarrow \infty} \frac{S_n}{T_n} = \frac{1}{3 \times \lambda}$ then find λ .
51. Let $f(x) = ax^2 + bx + c$, where $a, b, c \in \mathbb{R}$ and $f^2(1) + f^2(2) + f^2(3) - 2f(1) - 4f(2) - 6f(3) + 14 = 0$ and If $g(x) = \begin{vmatrix} f(x) & f(x+1) & f(x+2) \\ f'(x) & 2f'(x+1) & 3f'(x+2) \\ (f(x))^2 & (f(x+1))^2 & (f(x+2))^2 \end{vmatrix}$ then find $g'(1) + g(1)$: Where f' and g' represents derivatives.
52. If $f(\theta) = (\sec \theta + \operatorname{cosec} \theta)(\sin \theta + \cos \theta) - \sec \theta \operatorname{cosec} \theta$ lies completely between the roots of the quadric equation $(a-2)x^2 + 2ax + a + 8 = 0$ for all permissible values of θ , then find the number of integer values of a .
53. If $f(x) = \lim_{n \rightarrow \infty} \left(\sum_{r=1}^n r \cos^r x \right)$, then the value of $\left[\lim_{x \rightarrow 0} \left((1 - \cos x)^2 f(x) \right)^{\frac{1}{\cos x - 1}} \right]$ is (where $[\cdot]$ is greatest integer function)
54. If f and g are two positive valued functions such that $\lim_{x \rightarrow 2} f(x) = \infty$ and $\lim_{x \rightarrow 2} \frac{f^2(x)}{g(x)} = 0$, then $L = \lim_{x \rightarrow 2} \frac{\sqrt{16f^6(x) + 4g^3(x) + 5} + \sqrt[3]{f^8(x) + g^4(x) + 3}}{\sqrt[4]{f^{12}(x) + g^6(x) + 2} + \sqrt[5]{f^{12}(x) + g^6(x) + 1}}$ equals.....





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RANK

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37 SUMEDH SS HT.No. 231029088	38 V DEERAJ SATHVIK REDDY HT.No. 236067151	44 V VERGIYA KAUSHAL HT.No. 231004058	46 SUTHAR HARSHAL HT.No. 231005060	58 ABHI JAIN HT.No. 232091036	61 RIDAM JAIN HT.No. 232068021	62 SUTHAR HARSH HT.No. 232100028	63 SOJITRA DIPEN HT.No. 231070001	65 D PANEENDRANADHA REDDY HT.No. 236050298
67 GUNDA SUSHANTH HT.No. 236129732	68 HARSH TAYA HT.No. 237021718	69 KRISH GUPTA HT.No. 232068189	73 KUNCHAKURTHY VIVEK HT.No. 236134093	81 N DHARMA TEJA REDDY HT.No. 236050138	84 S VENKAT KOUNDINYA HT.No. 236130679	87 MRUNAL SHRIKANT HT.No. 231041138	92 SAMOTA APURVA HT.No. 232099270	97 N PRINCE BRANHAM REDDY HT.No. 236064172

BELOW
20
All India Open
Category Ranks

10
RANKS (50%)

BELOW
100
All India Open
Category Ranks

32

BELOW
1000
All India Open
Category Ranks

181

BELOW
100
All India
Category Ranks
Count

89

BELOW
1000
All India
Category Ranks
Count

699

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